

# A Scalar Sharp Kernel Whose Coordinate Multiplier Is Not Weakly Homogeneous

Candidate full solution packet  
arXiv:1811.05428, Section 3, PDF page 8

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**Status.** Candidate full affirmative answer, subject to human review. The construction is elementary and appears not to require the matrix-valued jet machinery used in the source paper.

## 1 Source question

Ghara [1, Section 3, p. 8] notes that examples are known for matrix-valued sharp kernels, but leaves open the scalar case:

### 3. WEAKLY HOMOGENEOUS OPERATORS IN THE CLASS $\mathcal{FB}_2(\mathbb{D})$

Although there are examples (see [17] Theorem (1.1)'], [19] Theorem 3.3]) of scalar valued sharp kernels  $K$  such that the composition operators  $C_\varphi$ ,  $\varphi \in \text{Möb}$ , are not bounded on  $(\mathcal{H}, K)$ , it does not necessarily follow that the multiplication operator  $M_z$  on  $(\mathcal{H}, K)$  fails to be weakly homogeneous. In many other examples excluding the ones in [17] and [19], the operator  $C_\varphi$  is bounded for all  $\varphi$  in  $\text{Möb}$  showing that the corresponding multiplication operator  $M_z$  is weakly homogeneous. While the question of the existence of an operator  $M_z$  which is not weakly homogeneous on a Hilbert space  $(\mathcal{H}, K)$ , where  $K$  is a scalar valued sharp kernel, remains unanswered, in this section, we find such examples where the kernel  $K$  takes values in  $\mathcal{M}_2(\mathbb{C})$ .

In the terminology of the source, a scalar positive-definite kernel  $K$  on  $\mathbb{D}$  is *sharp* when the coordinate multiplier  $M_z$  is bounded and

$$\ker(M_z^* - \bar{w}I) = \text{span}\{K(\cdot, w)\} \quad (w \in \mathbb{D}).$$

The question asks whether such a kernel can have  $M_z$  not weakly homogeneous. Recall that weak homogeneity requires  $\varphi(M_z)$  to be similar to  $M_z$  for every disk automorphism  $\varphi$ .

## 2 Idea of the proof

Start with the Hardy space, but adjoin one holomorphic function having a single boundary pole:

$$g(z) = \frac{1}{1-z}, \quad \mathcal{H} = H^2 \dot{+} \mathbb{C}g.$$

The Hardy space controls all eigenvectors over the open disk, so the resulting scalar kernel remains sharp. The identity  $zg = g - 1$  makes multiplication by  $z$  bounded and gives an explicit triangular  $2 \times 2$  block matrix.

The added one-dimensional quotient leaves a detectable spectral fingerprint:  $M_z^*$  acquires the boundary eigenvalue 1, and no other boundary eigenvalue. A nontrivial rotation moves this exceptional eigenvalue around the circle. Since similarity preserves the point spectrum of the adjoint,  $M_z$  cannot be similar to all its rotations.

### 3 Construction and proof

Let  $H^2 = H^2(\mathbb{D})$  with Szegő kernel  $k(z, w) = (1 - z\bar{w})^{-1}$ , and put

$$g(z) := \frac{1}{1 - z}.$$

Because  $g \notin H^2$ , every element of the algebraic sum

$$\mathcal{H} := H^2 \dot{+} \mathbb{C}g$$

has a unique representation  $h + cg$ . Make this an orthogonal Hilbert sum:

$$\|h + cg\|_{\mathcal{H}}^2 := \|h\|_{H^2}^2 + |c|^2.$$

Point evaluation is bounded on each summand. Thus  $\mathcal{H}$  is a scalar reproducing-kernel Hilbert space with kernel

$$K(z, w) = \frac{1}{1 - z\bar{w}} + \frac{1}{(1 - z)(1 - \bar{w})}. \quad (1)$$

The kernel is strictly positive definite: its first summand is the strictly positive-definite Szegő kernel, and the second is positive semidefinite.

Let  $T = M_z$  on  $\mathcal{H}$ , and identify  $\mathcal{H}$  unitarily with  $H^2 \oplus \mathbb{C}$  by  $h + cg \leftrightarrow (h, c)$ . Since  $zg = g - 1$ ,

$$T(h, c) = (Sh - c\mathbf{1}, c), \quad T^*(h, c) = (S^*h, c - \langle h, \mathbf{1} \rangle), \quad (2)$$

where  $S = M_z$  on  $H^2$  and  $\mathbf{1}$  is the constant function. In particular,  $T$  is bounded. Its upper-triangular block form has diagonal entries  $S$  and 1, hence

$$\sigma(T) = \overline{\mathbb{D}}. \quad (3)$$

**Lemma 1.** *The point spectrum of  $T^*$  is*

$$\sigma_p(T^*) = \mathbb{D} \cup \{1\}.$$

Moreover, for every  $w \in \mathbb{D}$ ,

$$\ker(T^* - \bar{w}I) = \text{span}\{K(\cdot, w)\}.$$

*Proof.* Write  $k_w(z) = (1 - z\bar{w})^{-1}$ . The standard shift calculation gives  $S^*k_w = \bar{w}k_w$ , and each eigenspace of  $S^*$  over  $\mathbb{D}$  is one-dimensional. Solving

$$T^*(h, c) = \bar{w}(h, c)$$

in (2) first gives  $h = ak_w$ . Since  $\langle k_w, \mathbf{1} \rangle = 1$ , the scalar equation then gives

$$c = \frac{a}{1 - \bar{w}}.$$

Under the identification  $\mathcal{H} = H^2 \oplus \mathbb{C}$ , the reproducing vector from (1) is precisely

$$K(\cdot, w) \longleftrightarrow \left( k_w, \frac{1}{1 - \bar{w}} \right).$$

This proves the kernel identity and sharpness over  $\mathbb{D}$ .

For a general eigenvalue  $\lambda$ , the first component in (2) requires  $S^*h = \lambda h$ . If  $|\lambda| < 1$ , the same calculation supplies one eigenvector. If  $|\lambda| = 1$ , then  $h = 0$ , and the second component is  $c = \lambda c$ ; it has a nonzero solution exactly when  $\lambda = 1$ . There are no eigenvalues outside  $\overline{\mathbb{D}}$ . This proves the asserted point spectrum.  $\square$

**Theorem 2.** *The kernel  $K$  in (1) is a scalar-valued sharp positive-definite kernel on  $\mathbb{D}$ , its coordinate multiplier is bounded, and that multiplier is not weakly homogeneous. Consequently the scalar sharp-kernel question in [1, Section 3, p. 8] has an affirmative answer.*

*Proof.* Only weak homogeneity remains to be excluded. Fix  $\theta \notin 2\pi\mathbb{Z}$  and take the disk rotation  $\rho_\theta(z) = e^{i\theta}z$ . Functional calculus gives  $\rho_\theta(T) = e^{i\theta}T$ . Lemma 1 yields

$$\sigma_p(\rho_\theta(T)^*) = e^{-i\theta}\sigma_p(T^*) = \mathbb{D} \cup \{e^{-i\theta}\},$$

whereas  $\sigma_p(T^*) = \mathbb{D} \cup \{1\}$ . The two sets differ.

If  $T$  and  $\rho_\theta(T)$  were similar, their adjoints would also be similar, and similar operators have identical point spectra. This contradiction shows that  $T$  is not similar even to this rotation of itself, so it is not weakly homogeneous.  $\square$

## 4 Verification and novelty bounds

The proof is checkable from the two displayed block equations. In particular: (i)  $g \notin H^2$  makes the direct sum unambiguous; (ii)  $zg = g - 1$  proves invariance and boundedness; (iii) the complete eigenvalue equations for  $T^*$  prove both sharpness and the exceptional boundary point; and (iv) a single rotation is enough to refute weak homogeneity.

A bounded novelty search covered the four run indexes, the local arXiv source corpus, web/arXiv exact-phrase searches for “scalar-valued sharp kernel” and “not weakly homogeneous”, the source title and arXiv id, and searches for the specific space  $H^2 \dot{+} \mathbb{C}(1 - z)^{-1}$ . It found the source paper and the author’s thesis restating the problem, but no later answer or occurrence of this construction. This is evidence only, not a guarantee of novelty.

Two other extraction signals in the source do not affect Theorem 2. The headline Bagchi–Misra question is answered in the source itself. The stated Shields power-growth conjecture is known to be false, as discussed by Bonilla and Müller [2].

**Human-review recommendation.** High priority: the example is short and settles the literal scalar-kernel question. A reviewer should confirm that the source intends exactly its stated definition of “sharp”, rather than an unstated Cowen–Douglas surjectivity hypothesis, and should independently repeat the novelty search.

## References

- [1] S. Ghara, *The orbit of a bounded operator under the Möbius group modulo similarity equivalence*, arXiv:1811.05428; Israel J. Math. 238 (2020), 167–207.
- [2] A. Bonilla and V. Müller, *Kreiss bounded and uniformly Kreiss bounded operators*, arXiv:1912.07931.